where β is fitted for as free parameter. We have defined the map using the 230 MHz GSM, although the 408 MHz map is an equally appropriate choice and produces very little difference in results.

The Bayesian evidence and posterior samples of the foreground and signal models are found using PolyChord, with uniform priors in the range $2.45844 < \beta < 3.14556$, the full range of a spectral index map derived using the 230 MHz and 408 MHz GSMs (De Oliveira-Costa et al. 2008; Anstey et al. 2021). Testing of the pipeline is done using simulated data which is calculated using the 230 MHz GSM and the spectral index map. A Gaussian mock 21-cm global signal of the form

$$T_{\rm sg}(\nu) = -A_{21} \exp\left(-\frac{(\nu-\nu_c)^2}{2\sigma_{21}^2}\right), \tag{10}$$

where A_{21} is the signal amplitude, ν_c, the centre frequency and σ_{21}, the signal width, was added to the simulated data. As this is a similar shape to the physical 21-cm signal, this is a suitable analogue for testing purposes and is vastly less computationally expensive to model than other more physically motivated models. The signal parameters have uniform priors in the ranges $50 < \nu_c < 200$ MHz, $10 < \sigma_{21} < 20$ MHz and $0 < A_{21} < 0.25$ K. When testing the pipelines, they will be run with a 21cm signal with parameters $A_{21} = 0.155$ K, $\sigma_{21} = 15$ MHz and $\nu_c = 80$ MHz. Gaussian noise with standard deviation $\hat{\sigma} = 0.1$ K is added to the data to simulate the uncorrelated noise of the system.

The REACH pipeline's method of fitting of the foreground parameter can make use of time dependent data in the pipeline (Anstey et al. 2022). The observation is split into N_t consecutive integrations, or time bins, which are measured in local sidereal time (LST) to match the observation to the stage of Earth's rotation. We hence define the likelihood like so,

$$\log \mathcal{L}_{\rm std} = \sum_i \sum_j -\frac{1}{2}\log(2\pi\sigma_{0,\rm std}^2) - \frac{1}{2}\left(\frac{T_{\rm data}(\nu_i,t_j) - (T_{\rm fg}(\nu_i,t_j) + T_{21}(\nu_i) + T_{\rm CMB})}{\sigma_{0,\rm std}}\right)^2, \tag{11}$$

where i refers to the ith frequency bin, and j to the jth time bin. This likelihood will be referred as the 'standard pipeline' from here on.

2.3 Systematic Model

As discussed in section 1.1, we might expect to find systematics in the REACH system which are sinusoidal, generated by standing waves in the receiver or on the antenna. As a result we can model the general systematic we may expect to see as a damped sinusoid of the form

$$T_{\rm sys}(\nu) = A_{\rm sys}\left(\frac{\nu}{\nu_{0,\rm sys}}\right)^{-\alpha_{\rm sys}} \sin\left(\frac{2\pi\nu}{P_{\rm sys}} + \phi_{\rm sys}\right), \tag{12}$$

where $\nu_{0,\rm sys} = 50$ MHz is the fiducial radio frequency of the systematic, $A_{\rm sys}$ is the amplitude of the systematic, $P_{\rm sys}$ is the period of the systematic, $\phi_{\rm sys}$ is the phase of the systematic, and $\alpha_{\rm sys}$ is the dampening of the systematic (Scheutwinkel et al. 2022). In this paper the value of the dampening is fixed at $\alpha_{\rm sys} = 1.4$.

The model of the time-varying systematic we will consider in this paper is the case where a systematic is constant in phase, frequency and dampening but modulates its amplitude according to the incoming power from the galactic foreground. Here we define the amplitude of the systematic at time bin j as

$$A_{\rm sys}(t_j) = A_{\rm sys}(t_0) \cdot \frac{T_{\rm fg}(\nu=\nu_0, t=t_j)}{T_{\rm fg}(\nu=\nu_0, t=t_0)}, \tag{13}$$

where ν_0 determines the radio frequency from which the foreground power is taken. Here, we will take $\nu_0 = 50$ MHz. Currently the modulation is done monochromatically although it may be better to model the systematic by modulating each frequency bin separately in future.

Figure 4 shows this foreground-modulated systematic for 24 time bins of length 15 minutes. Here, the systematic shows a slight shift in amplitude as the foreground power increases over time. As there is no change in the phase or frequency of the systematic over time, the signal does not average down by any significant amount and as such its amplitude cannot be affected by increasing integration times.

2.4 Gaussian Processes

In order to account for the covariance in the model residuals introduced by the presence of a systematic in the data, we will use Gaussian processes (GPs) to build upon the standard REACH likelihood. Gaussian processes are non-parametric probabilistic methods of performing regression and forecasting that have found particular use for Bayesian time series regression (MacKay 2002; Roberts et al. 2013).

We define a GP as a collection of random variables which have consistent joint Gaussian distributions (Rasmussen 2004). These Gaussian distributions are defined by the covariance function, or kernel, of the Gaussian process where the choice of kernel is arbitrary, depending on the data you are trying to model. There is a wealth of literature into the variety of structures of kernel that can be used for GP regression, where their applicability depends on the problem that is trying to be solved.

The kernel, which we will use in this paper is the squared exponential,

$$K_{\rm SE}(t_i,t_j) = \sigma_{\rm SE}^2 \exp\left(-\frac{|t_i-t_j|^2}{2\ell^2}\right), \tag{14}$$

where ℓ is known as the characteristic length scale of the Gaussian process and $\sigma_{\rm SE}^2$ is the scale factor of the squared exponential kernel (Rasmussen & Williams 2006).

We choose this kernel as it has a simple form and describes a family of smooth functions, as seen in figure 3, of the form which we expect the systematic to take – particularly for the smoothly modulated systematics simulated here. There are many other kernel choices which are valid, for example a periodic kernel (Rasmussen & Williams 2006) which will be useful which the systematic is modulated by the periodic galactic foreground power. We could also introduce a 2D Gaussian process kernel to incorporate correlations between frequency bins.

The covariance matrix we construct is hence

$$\mathbf{C}_{ij} = K(t_i,t_j) = \sigma_{0,\rm GP}^2 + K_{\rm SE}(t_i,t_j), \tag{15}$$

where $\sigma_{0,\rm GP}$ is the Gaussian signal noise and is equivalent to adding a white noise kernel to the squared exponential kernel. We set the prior on $\sigma_{0,\rm GP}$ to be a log uniform prior in the range $10^{-4} \leq \sigma_{0,\rm GP} \leq 0.5$ K, on $\sigma_{\rm SE}$ to be a log uniform prior in the range $0.01 \leq \sigma_{\rm SE} \leq 0.5$ K and on ℓ to be a uniform prior in the range $100 \leq \ell \leq 1000$ minutes. The prior on the uncorrelated noise is taken from the standard REACH pipeline (Anstey et al. 2021) while the scale factor and characteristic length priors are informed by the amplitude and time variance respectively of the foreground-modulated systematic we insert into the data. This covariance matrix is then combined with equation 5, which will be referred to as the likelihood for the 'Gaussian process pipeline', $\mathcal{L}_{\rm GP}$.

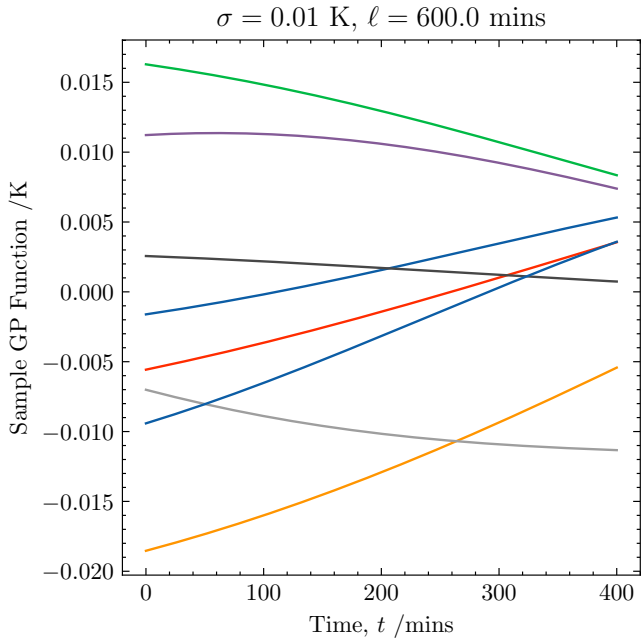


Figure 3. Example of eight functions drawn from the squared exponential Gaussian Process kernel given by equation 14, with a scale factor of 0.01 K and a covariance length of 600 minutes.

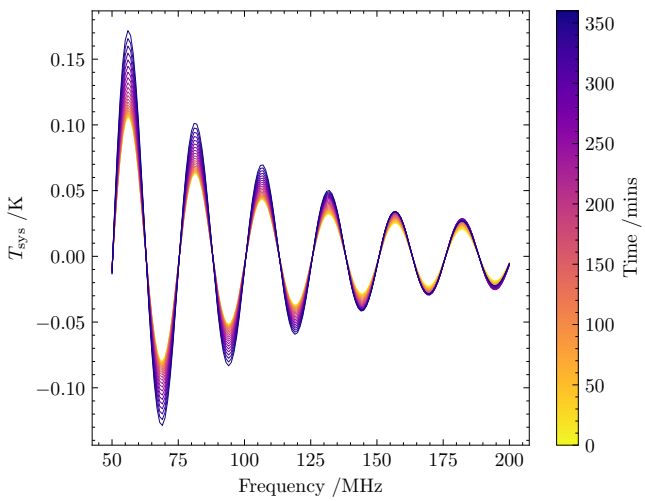


Figure 4. Example of a damped sinusoidal systematic given by equations 12 and 13 which is modulated by the incoming power from the galactic foreground over 24 time bins of length 15 minutes. The initial amplitude of the systematic was set to $A_{\rm sys} = 0.209$ K.

Once the weighted mean hyperparameters, $\{\sigma_{0,\rm GP}, \sigma_{\rm SE}, \ell\}$, have been found using PolyChord and Anesthetic (Handley 2019), the mean regression line for a set of predicted times, $\mathbf{t}_{\rm pred}$, using the observed data, $\{\mathbf{t}_{\rm data}, \mathbf{T}_{\rm data}\}$, can be found as

$$\mu(\mathbf{t}_{\rm pred}) = K(\mathbf{t}_{\rm pred}, \mathbf{t}_{\rm data}) K(\mathbf{t}_{\rm data}, \mathbf{t}_{\rm data})^{-1} \mathbf{T}_{\rm data}, \tag{16}$$

and the covariance matrix of the predicted data given by

$$\begin{aligned}\mathbf{C}_{\rm pred} =& K(\mathbf{t}_{\rm pred}, \mathbf{t}_{\rm pred}) \\ &- K(\mathbf{t}_{\rm pred}, \mathbf{t}_{\rm data}) K(\mathbf{t}_{\rm data}, \mathbf{t}_{\rm data})^{-1} K(\mathbf{t}_{\rm data}, \mathbf{t}_{\rm pred}).\end{aligned} \tag{17}$$

3 RESULTS

In this section we will demonstrate the results of injecting a simulated sinusoidal systematic into both the standard and Gaussian process pipelines and compare the results of the two. In section 3.1 we will demonstrate the improvements to recovery of the 21cm signal made by the GP pipeline, in section 3.2 we will see how systematics with different parameters affect the standard and GP pipelines, and in section 3.3 we will demonstrate a potential secondary use of the GP pipeline for regression of the time variation of the model residuals.

3.1 Signal Recovery

We first added a foreground modulated systematic over 24 time bins of length 15 minutes with an initial amplitude of $A_{\rm sys} = A_{21} = 0.155$ K, period of $P_{\rm sys} = 0.5\sigma_{21} = 7.5$ MHz and phase of $\phi_{\rm sys} = \pi$. For the chosen time period the foreground modulated systematic amplitude varies monotonically, increasing to a final amplitude of $\sim 1.7A_{\rm sys}$. The effects of adding a systematic to the data on the ability of the standard and GP pipelines to recover the signal can be seen in figure 5. Plotted in green is the true signal and the blue contours show the 1, 2 and 3σ contours of the signal posterior plotted with fgivenx (Handley 2018). It can be seen that while the standard pipeline misses the signal by greater than 3σ, the GP pipeline has a much wider posterior, resulting in the pipeline capturing the signal within 1σ.

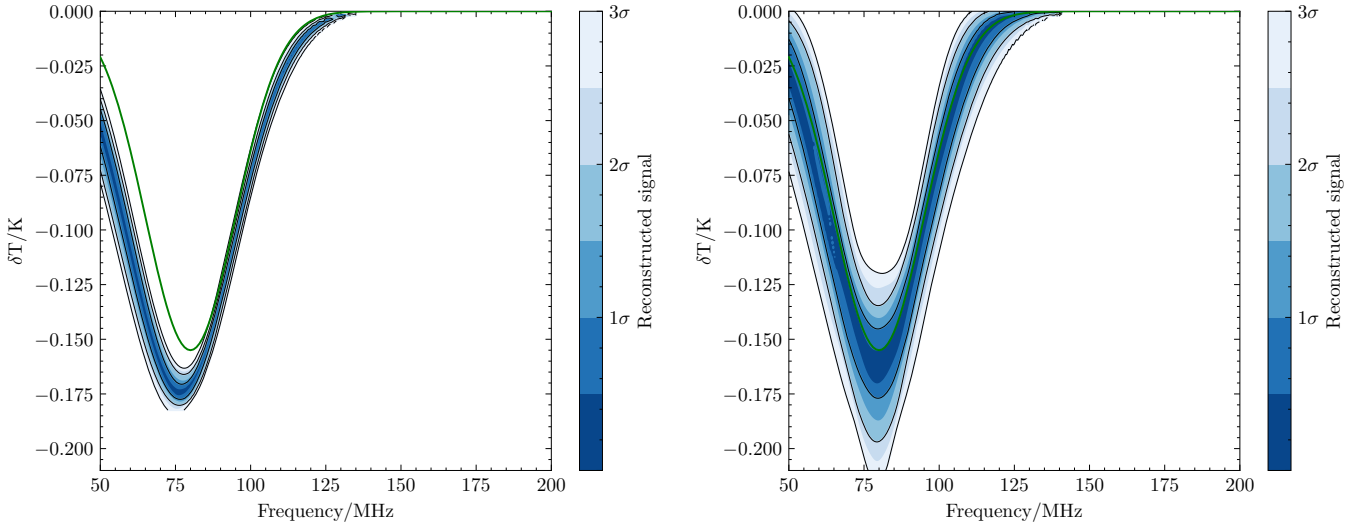
For the standard pipeline the mean noise parameter was $\sigma_{0,\rm std} = 0.0637 \pm 0.0008$ K. In the case of the GP pipeline, the mean hyperparameters were $\sigma_{0,\rm GP} = 0.0234 \pm 0.0003$ K, $\sigma_{\rm SE} = 0.064 \pm 0.004$ K and $\ell = 640 \pm 60$ minutes. This shows the ability of the GP pipeline to separate the uncorrelated Gaussian noise from the correlated systematic, something which could not be done with a standard Gaussian likelihood which absorbs both noise and systematic in the $\sigma_{0,\rm std}$ parameter.

When no systematic is added to the data both the standard and GP pipelines recovered the signal parameters to within 1σ. For the standard pipeline the mean noise parameter was $\sigma_{0,\rm std} = 0.0245 \pm 0.0003$ K. In the case of the GP pipeline, the mean hyperparameters were $\sigma_{0,\rm GP} = 0.0245 \pm 0.0003$ K, $\sigma_{\rm SE} = 0.0101 \pm 0.0009$ K and $\ell = 500 \pm 200$ minutes. In this case the posterior of the $\sigma_{\rm SE}$ parameter has saturated at the lower end of its prior and can be assumed to either be very small or zero. Combining this with the fact that $\sigma_{0,\rm std} = \sigma_{0,\rm GP}$, this shows that the GP pipeline can be an important diagnostic tool to identify unmodelled systematics in the data, with the amplitude of the time-correlated noise parameter falling to zero in the absence of a systematic.

3.2 Varying Systematic Parameters

To test the limits of the robustness of the standard and GP pipelines to systematics we varied the parameters of the simulated systematic to see how the goodness-of-fit and parameter biases changed. We repeated the pipeline runs with systematics with initial amplitudes of $A_{\rm sys}/A_{21} = \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00\}$, periods of $P_{\rm sys}/\sigma_{21} = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ and phases of $\phi_{\rm sys} = \{0, 0.5\pi, \pi, 1.5\pi\}$. For each of these parameter combinations, both the standard and GP pipelines were run with time-separated data with 24 time bins of length 15 minutes – equivalent to a single nights' observation.

We judge the goodness-of-fit of the pipeline fit using a root-mean-



(a) Standard Pipeline

(b) GP Pipeline

Figure 5. Comparison of the recovered signal posteriors plotted with FGIVENX (Handley 2018) for the standard pipeline (equation 11) and the Gaussian process pipeline (equations 5 and 15). In green is plotted the 'true' signal which was added into the simulated data. There is a foreground modulated systematic with an initial amplitude of $A_{\rm sys} = A_{21} = 0.155$ K, period of $P_{\rm sys} = 0.5\sigma_{21} = 7.5$ MHz and phase of $\phi_{\rm sys} = \pi$ present in the data.

square error (RMSE) value. This is calculated as

$$\mathrm{RMSE} = \left(\frac{\sum_i \sum_j w_j \left[T_{\rm sg}(\nu_i, \theta^*_{{\rm sg},j}) - T_{\rm sg}(\nu_i, \theta_{\rm true,\,sg}) \right]^2}{N_\nu \sum_j w_j} \right)^{\frac{1}{2}}, \quad (18)$$

where $\theta^*_{{\rm sg},j}$ and w_j are the signal posterior samples and their weights respectively, N_ν is the number of frequency bins and $\theta_{\rm true,\,sg}$ are the true signal parameters.

Figure 6 shows the root mean square error for the standard pipeline (lower row) and the Gaussian process pipeline (upper row). It can be seen in most cases that the GP pipeline improves the goodness-of-fit (lower RMSE). This can be attributed to a combination of the GP pipeline widening the signal posterior whilst also reducing the biasing on the fitted signal parameters when a systematic is present in the data.

To compare how the confidence in the ability of the pipeline to detect the global signal, we use the Bayes factor

$$\log \mathcal{K} = \log \mathcal{Z}_{\rm GP} - \log \mathcal{Z}_{\rm std}, \quad (19)$$

where $\mathcal{Z}_{\rm GP}$ is the Bayesian evidence given when the GP pipeline is run, and $\mathcal{Z}_{\rm std}$ is when the standard pipeline is run. This factor gives us the odds, $1 : \mathcal{K}$, that the data prefers GP pipeline over the standard pipeline (MacKay 1991).

We see in figure 7, for all phases and most periods and amplitudes of the sinusoidal systematic, the Bayes factor is equal to or exceeds 20 indicating that the Gaussian process pipeline is highly favoured over the standard pipeline. In particular we find a minimum $\log \mathcal{K}$ value of 4.8, corresponding to minimum betting odds of around $1 : 120$ in favour of the GP pipeline's. We consider a Bayes factor over 2.5 to be a significant favouring and a Bayes factor over 5 to be a decisive favouring in line with the guidelines given by Jeffreys (1998). In the absence of a simulated systematic $\log \mathcal{K} = -64.0$, indicating a decisive preference for the standard pipeline over the GP pipeline – a difference which may be attributed to the Occam penalty of the Bayesian evidence penalising the extra two GP hyperparameters (Hergt et al. 2021).

We now look at the individual signal parameters to more clearly see how the widening posterior alongside the reduction in bias affects the error in the parameter estimation. Figures 8, 9 and 10 show the error in the recovery of the signal centre frequency, ν_c, signal amplitude, A_{21}, and signal width σ_{21}, respectively for different systematics. It can be seen that there is a marked improvement in the recovery of the signal parameters when using the GP pipeline. In particular, when the systematic period is smaller than twice the signal width the GP pipeline is able to recover the parameters to within 2σ in most cases although has difficulty recovering ν_{21} when $\phi_{\rm sys} = 0.5\pi$ as the first trough of the systematic is close to the true signal centre frequency. For larger systematic periods there is a slight improvement in the RMSE but still miss the signal parameters by over 2σ, an effect which is likely caused by the troughs of the long-period sinusoids more closely representing the true Gaussian signal. In general varying the systematic amplitude has only a slight effect on the Gaussian process pipeline when the systematic period is low.

In order to discover the limits of the Gaussian process pipeline when run with large systematics, we also repeat the same parameter sweep with much greater systematic amplitudes of $A_{\rm sys}/A_{21} = \{4.0, 8.0, 12.0, 16.0\}$ and periods of $P_{\rm sys}/\sigma_{21} = \{1.0, 2.0, 3.0, 4.0\}$. The RMSE values of these systematics are shown in figure 11. It can be seen that when the systematic amplitude four times the signal amplitude or greater, the GP pipeline no longer provides an improvement and in some cases worsens the RMSE value. Analysis of the posterior distributions of the hyperparameters shows that for the largest amplitude systematics the correlated noise hyperparameter, $\sigma_{\rm SE}$, saturates at the higher end of its prior, demonstrating the need to adjust the original priors should there be a very large systematic in the data.

Furthermore we can test the pipelines in the case where there is no global signal in the data but a Gaussian signal model is still being fitted for. Here we test the pipelines for systematics with amplitudes $A_{\rm sys}/(0.155\ \mathrm{K}) = \{1.00, 2.00, 3.00\}$ and periods $P_{\rm sys}/(15\ \mathrm{MHz}) = \{1.0, 2.0, 3.0\}$. Figure 12 shows the error in the fitted signal amplitude, where the true amplitude is 0 K. While the fit worsens somewhat at a phase of 0.5π for low periods, in general there is an improvement in the ability to identify a lack of global signal in